Evaluating the Generalizability of Heart Disease Prediction Models Across Heterogeneous Populations

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Information Table:

Reference (Year)	***Description***	***Advantages***	***Disadvantages***	***References***
BMC Medical Research Methodology (2025)	Discusses statistical considerations for validating predictive models across heterogeneous medical populations and emphasizes rigorous external validation methods.	- Provides methodological guidance for validating clinical prediction models. - Highlights the importance of cross-population evaluation.	- Focuses more on methodology than algorithm development.	https://link.springer.com/article/10.1186/s12874-025-02463-y
JAMA Internal Medicine (2021)	Evaluates machine learning models used in clinical risk prediction and compares them with traditional statistical approaches.	- Demonstrates improved discrimination of ML models. - Large-scale clinical evaluation.	- Limited discussion of distribution shift across institutions.	https://jamanetwork.com/journals/jamainternalmedicine/fullarticle/2781307

Scientific Reports (2022)	Applies machine learning algorithms such as random forests and gradient boosting to structured medical datasets for disease prediction.	- Shows strong predictive performance of ML classifiers. - Demonstrates applicability to clinical data.	- Validation primarily performed on internal datasets.	https://www.nature.com/articles/s41598-022-12497-7
JAMA Cardiology (2021)	Investigates machine learning methods for cardiovascular risk prediction using large clinical datasets.	- Demonstrates improved prediction of cardiovascular outcomes. - Uses real-world clinical data.	- Limited focus on cross-population generalization.	https://jamanetwork.com/journals/jamacardiology/fullarticle/2777055
npj Digital Medicine (2021)	Examines how distribution shifts affect the reliability of machine learning models in healthcare systems.	- Highlights risks of dataset shift in medical AI. - Emphasizes importance of external validation.	- Mostly theoretical evaluation of deployment risks.	https://www.nature.com/articles/s41746-021-00549-7
npj Digital Medicine (2024)	Discusses monitoring and evaluation strategies for maintaining performance of AI models after	- Addresses model drift and real-world reliability. - Emphasizes long-term	- Focuses more on deployment than model development.	https://www.nature.com/articles/s41746-024-01085-w

	deployment in healthcare.	monitoring of AI systems.		
JMIR (2024)	Analyzes the application of AI-based predictive models within digital health systems and their clinical implications.	- Highlights benefits of ML for healthcare decision support. - Discusses implementation challenges.	- Limited experimental validation across institutions.	https://www.jmir.org/2024/1/e51409/
PMC10513895 (2023)	Evaluates predictive model robustness when applied across heterogeneous medical datasets.	- Demonstrates performance decline across datasets. - Supports need for external validation.	- Limited number of cohorts analyzed.	https://pmc.ncbi.nlm.nih.gov/articles/PMC10513895/
PMC8478611 (2021)	Multicenter validation study evaluating machine learning models in clinical environments.	- Shows impact of institutional variability. - Uses multiple clinical datasets.	- Requires recalibration when applied to new populations.	https://pmc.ncbi.nlm.nih.gov/articles/PMC8478611/
Frontiers in Digital Health (2025)	Discusses ethical and operational challenges of deploying AI diagnostic systems in healthcare.	- Addresses transparency and accountability of AI models. - Highlights risks of unvalidated deployment.	- Focuses more on policy and ethics than predictive modeling.	https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1403047/full

BMJ (2023)	Provides methodological guidance on evaluating and reporting clinical prediction models.	- Strong clinical research standards. - Emphasizes external validation and calibration.	- Focused mainly on reporting standards.	https://www.bmj.com/content/384/bmj-2023-074819
JAMA Network Open (2025)	Investigates performance variability of predictive models across healthcare systems.	- Demonstrates generalization issues in medical AI. - Uses large clinical datasets.	- Models may require retraining for new populations.	https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2834882
PMC9281065 (2022)	Reviews fairness and bias considerations in machine learning models used in healthcare decision-making.	- Addresses demographic bias in AI systems. - Emphasizes fairness evaluation.	- Mostly conceptual rather than experimental.	https://pmc.ncbi.nlm.nih.gov/articles/PMC9281065/
JAMIA Open (2025)	Evaluates cross-site performance of predictive models in healthcare institutions.	- Demonstrates importance of cross-site validation. - Uses real-world healthcare data.	- Limited dataset diversity in some experiments.	https://academic.oup.com/jamiaopen/article/8/5/ooaf115/8269328
European Heart Journal Digital Health (2024)	Explores machine learning applications in cardiovascular disease	- Focuses specifically on cardiovascular AI.	- Some studies rely on single-center datasets.	https://academic.oup.com/ehjdh/article/6/1/7/7845948

	prediction and risk stratification.	- Provides clinical insights for deployment.		
Annals of Internal Medicine (2023)	Evaluates the clinical performance of predictive models under external validation conditions.	- Demonstrates importance of model calibration. - Strong clinical study design.	- Limited exploration of algorithmic diversity.	https://www.acpjournals.org/doi/full/10.7326/M23-2960
American Journal of Cardiology (2024)	Investigates machine learning approaches for cardiovascular diagnosis and risk assessment.	- Demonstrates ML usefulness for cardiac prediction. - Uses structured cardiovascular datasets.	- Validation often limited to specific populations.	https://www.sciencedirect.com/science/article/pii/S0003496724085017
PMC12525484 (2024)	Examines fairness, bias, and heterogeneity in healthcare machine learning systems.	- Emphasizes demographic fairness evaluation. - Highlights risks of biased predictions.	- Mainly focuses on ethical implications rather than model design.	https://pmc.ncbi.nlm.nih.gov/articles/PMC12525484/
Journal of the American College of Cardiology (2024)	Investigates machine learning applications in cardiovascular risk prediction and clinical decision support.	- High-quality cardiovascular research. - Demonstrates strong predictive performance.	- Some models require extensive clinical data.	https://www.jacc.org/doi/abs/10.1016/j.jacc.2024.05.003

Dataset Reference:

Reference (Year)	***Description***	***Advantages***	***Disadvantages***	***Link***
Dua, D., & Graff, C. (2019)	A widely used dataset for heart disease prediction containing four cohorts: Cleveland, Hungary, Switzerland, and VA Long Beach with 13 clinical attributes.	- Standard benchmark dataset for cardiovascular prediction. - Contains heterogeneous patient populations.	- Some cohorts contain missing values.	https://archive.ics.uci.edu/dataset/45/heart+disease